

Saksham Bajpayi

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Education

PES University, Bengaluru 2024 – 2028
B.Tech in Computer Science Engineering

Experience

Open Source Contributor — PESU-Dev 2024
github.com/pesu-dev/auth

- Contributed to *pesu-auth*, a secure authentication API for validating PESU Academy credentials.
- Implemented backend authentication flows without storing user credentials, improving privacy and security.
- Collaborated via pull requests, code reviews, and documentation to improve developer usability.

Beta Tester — Pokemod 2021 – 2022
pokemod.dev

- Performed structured pre-production testing of a large-scale gaming application across multiple builds.
- Identified functional bugs, edge cases, and performance regressions in diverse scenarios.
- Worked closely with developers to validate fixes and improve application stability.

Projects

Explainable Visual Question Answering (VQA) System

github.com/SakshamBajpayi/VGA

- Built a multimodal AI system for answering natural-language questions on images using ViLT and BLIP-2.
- Designed a CLIP-based explainability module to compute patch-level similarity for region-level visual grounding.
- Generated interpretable outputs via heatmaps and bounding boxes to validate model predictions.
- Implemented an end-to-end pipeline integrating image encoding, cross-modal reasoning, and answer generation.

GPAI — Syllabus-Aligned RAG Study Assistant

- Developed a Retrieval-Augmented Generation pipeline using FAISS and local LLMs (Mistral-7B, Phi-Mini).
- Processed academic PDFs using custom pipelines for structured text extraction and semantic indexing.
- Enabled syllabus-aligned question answering with reduced hallucination through curated data sources.

Kairos — Intelligent Workflow Automation Platform

- Designed a natural-language-driven automation system for cross-application workflows.
- Built NLP pipelines to convert user intent into structured automation tasks.
- Integrated Gmail and Google Calendar using Node.js, MongoDB, Redis, and Socket.io for real-time automation.

Skills

Languages: Python, C/C++, Java, JavaScript, TypeScript, SQL, Verilog, Julia, Rust

Frameworks: React, Node.js, Express.js, Gatsby

ML / AI: PyTorch, TensorFlow, HuggingFace Transformers, scikit-learn

Databases: MongoDB, MySQL

Tools: Docker, REST APIs, OAuth 2.0, FAISS, NumPy, Pandas, Matplotlib, Vivado, Intel Quartis

Achievements

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- Top 10 among 50 teams at Arithmania Hackathon.
 - Cleared Stage 1 of RMO and NTSE.
 - Top 10 in KISA Talent Search Exam (State-level).
 - Top 20 nationally in a CTF competition among 180 teams.